

Interpretable and generative models for mouse spaceflight

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Biological & Physical Sciences

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PROJECT

One dataset, two machine-learning questions

We used mouse bulk RNA-seq to study how spaceflight changes genes, pathways, and tissue state.

NASA OSDR API Mus musculus bulk RNA-seq | spaceflight (FLT) and ground control (GC) | multiple missions and tissues

01 Which biological programs change?

expiMap

Train a pathway-constrained reference on ARCHS4, then map OSDR samples into mouse Reactome programs.

Output: tissue-specific pathway shifts

02 Can synthetic profiles sharpen the analysis?

Conditional generation

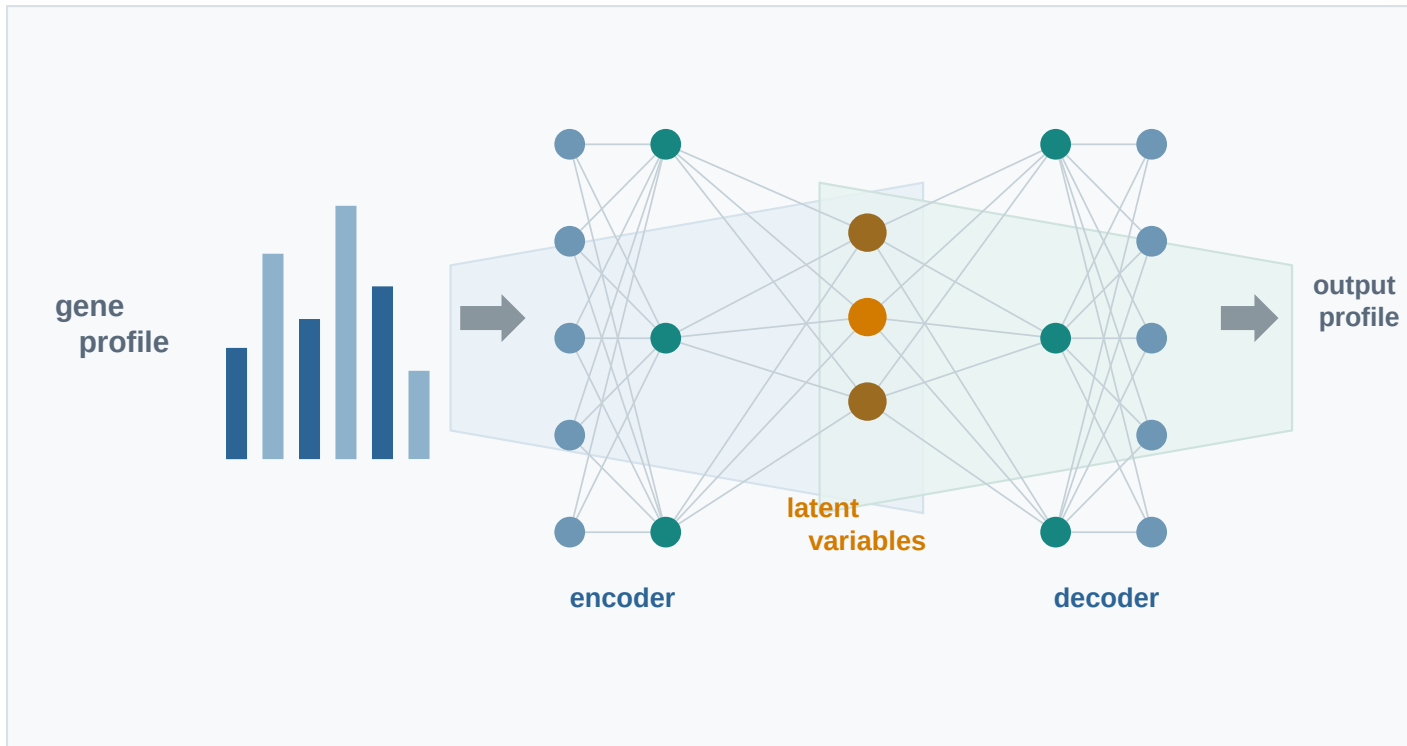
Generate matched expression profiles by tissue, study context, and FLT or GC condition.

Output: validated profiles and gene ranking

Autoencoders compress expression into a latent space

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



LATENT SPACE



expiMap: each constrained latent variable corresponds to a Reactome gene program.

EXPIMAP SCORES

Program scores turn expression into pathway-level changes

Each sample receives a Reactome program score; FLT minus GC gives the direction of change.

REPRESENTATIVE OBSERVED OSDR PROGRAMS					
		FLT score	-	GC score	= FLT - GC
Thymus	DNA repair	-0.45	-	+0.55	-1.01
Skin	Cell-cell junction organization	-0.09	-	+0.11	-0.20
Liver	MHC class II antigen presentation	-0.07	-	+0.09	-0.16
Spleen	T-cell receptor signaling	-0.13	-	+0.13	-0.26

lower

no shift

higher

HOW TO READ IT

- 1 Score each sample**
The constrained latent value represents one Reactome program.
- 2 Compare within project**
Flight samples are compared with ground controls from the same study.
- 3 Combine project shifts**
Project-level changes are integrated across the available missions.

Negative change = lower program score in flight

EXPIMAP RESULTS

Four tissues produced the clearest pathway patterns

Effects were estimated within each project before the project-level shifts were combined.

Thymus

117 samples | 5 projects

LOWER IN FLIGHT

DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions

Known involution may also involve lower repair, motility, and niche coordination.

Skin

151 samples | 4 projects

LOWER IN FLIGHT

Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions

Barrier injury may include a broader loss of tissue maintenance and coordination.

Liver

197 samples | 9 projects

LOWER IN FLIGHT

MHC class II antigen presentation; T-cell receptor signaling

Metabolic heterogeneity coexisted with lower adaptive immune communication.

Spleen

100 samples | 5 projects

LOWER IN FLIGHT

T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling

The most consistent multi-pathway result joined adaptive and innate immune changes.

BACKGROUND

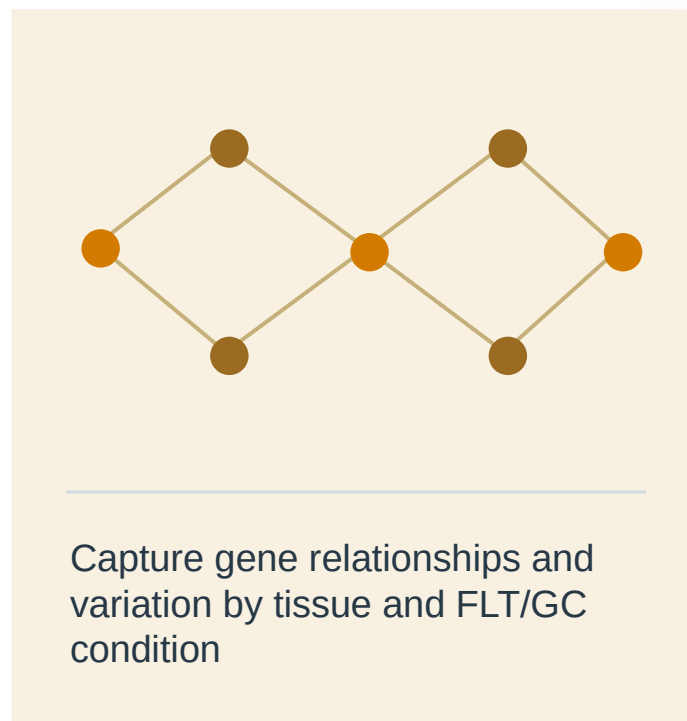
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



02 Learn the distribution



03 Synthetic profiles



QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610

profiles

75

accessions

835 FLT

775 GC



ARCHS4 mouse

Reference pretraining cohort

997,515

profiles screened



17,244

selected

20 tissues

Study effects can look like spaceflight biology.

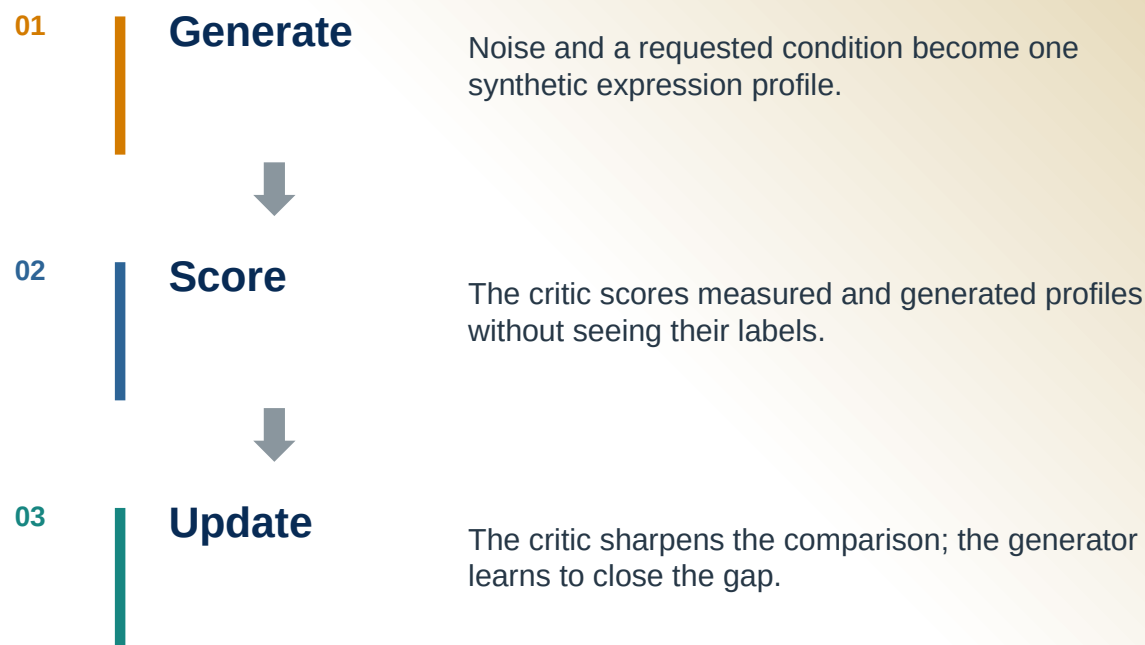
- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

A generator learns by competing with a critic

The generator makes conditioned expression profiles; the critic compares them with measured RNA-seq.



TRAINING LOOP



CONDITION

Tissue

FLT / GC

Accession

Material type

METHOD

We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

01 Data scope

Sources

OSDR API

ARCHS4

Studies

Single

Multiple

Tissues

Per tissue

All tissues

02 Transform

Expression

Raw

CPM

TPM

Scaling

None

Z-score

MaxAbs

Features

All

HVG

L1000

03 Harmonization

Global or study correction

None

Within-study z-score

ComBat / MBatch

MOBER

04 Model training

Generator

WGAN-GP

DDIM

Training source

OSDR only

ARCHS4 only

Pretrain + adapt

05 Conditioning

Model inputs

Tissue

FLT / GC

Accession

Material type

Sex / age

Downstream

configuration

PREPROCESS

TPM / MaxAbs / 974 landmarks

PRETRAIN

ARCHS4

ADAPT

OSDR

GENERATE

Conditional DDIM

CONDITION

Tissue

FLT / GC

Accession

Material type

HARMONIZE

None

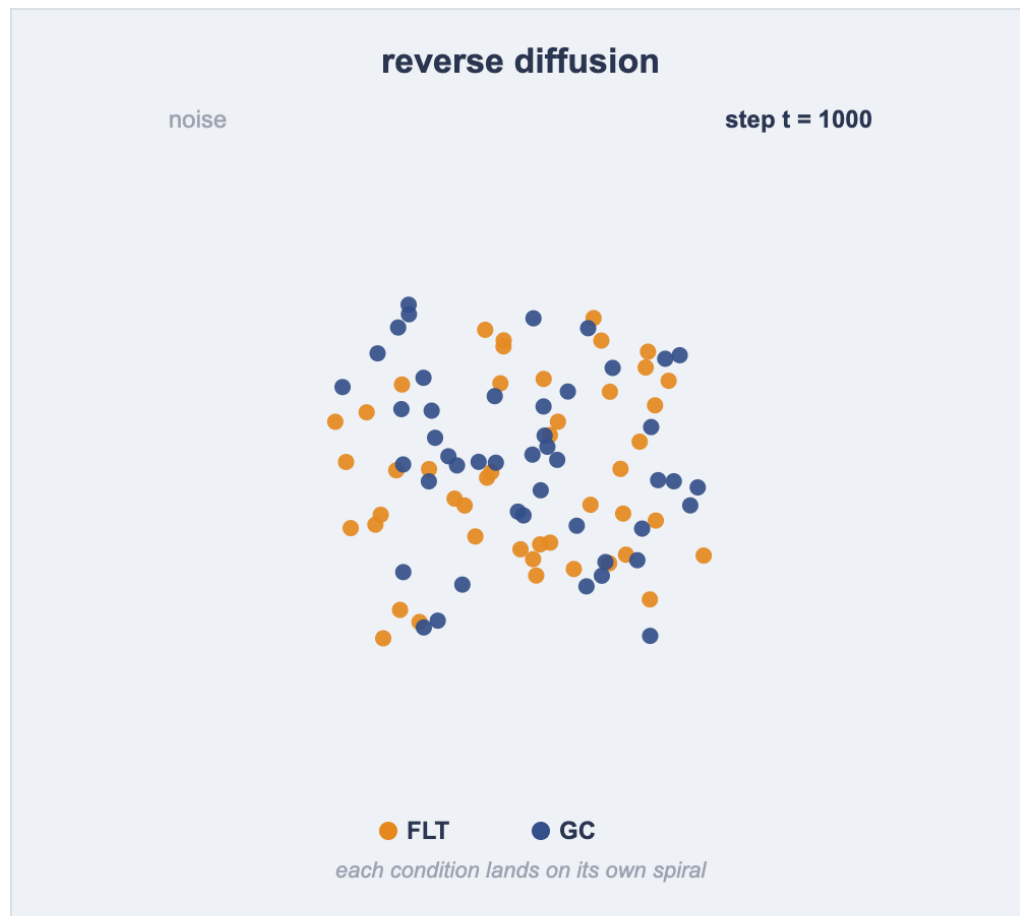
SCOPE

All tissues

DIFFUSION

Diffusion turns noise into a conditioned sample

The model repeatedly removes predicted noise while following the requested biological context.



FOR RNA-SEQ

$T = 1000$

Random noise

974 unconstrained gene values



REVERSE STEPS

Conditional denoising

Tissue

FLT / GC

Study

Material

$T = 0$

Synthetic expression

One generated 974-gene profile

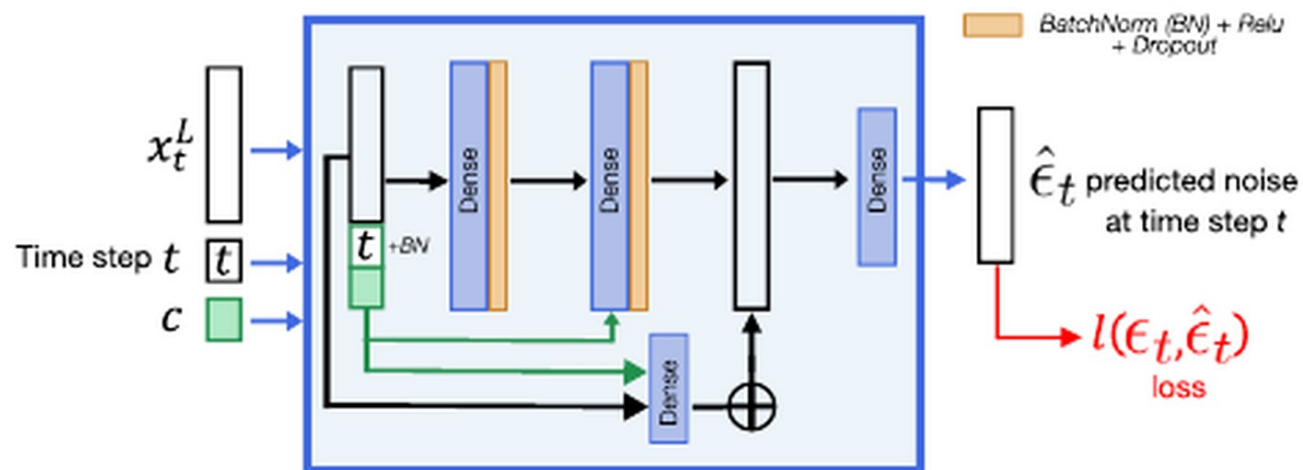


VALIDATION

DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

Architecture from Lacan et al.



Residual dense denoiser conditioned on timestep and tissue.

OSDR adaptation adds FLT/GC, accession and material context through LoRA.

ARCHS4 tissue probe

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial acc.	0.636	0.475	near 0.5
FD / real P95	0.144	0.074	lower

Use DDIM

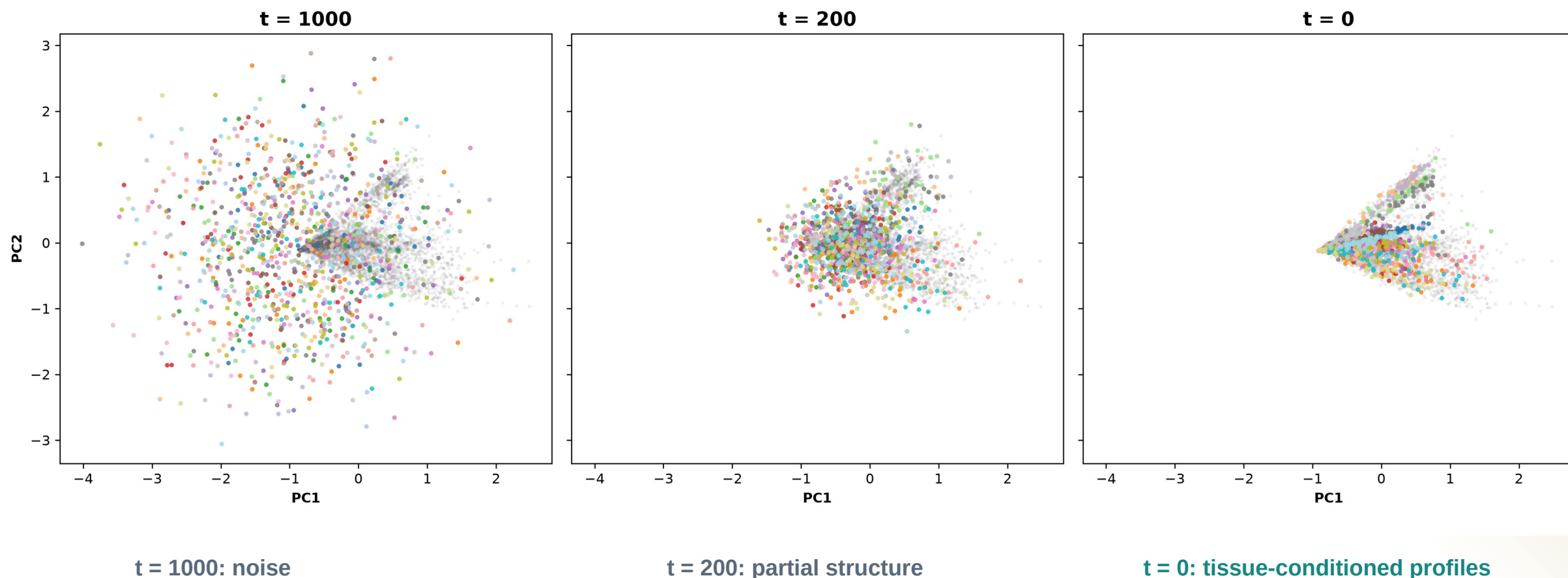
Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

DIFFUSION

Diffusion learns tissue structure from noise

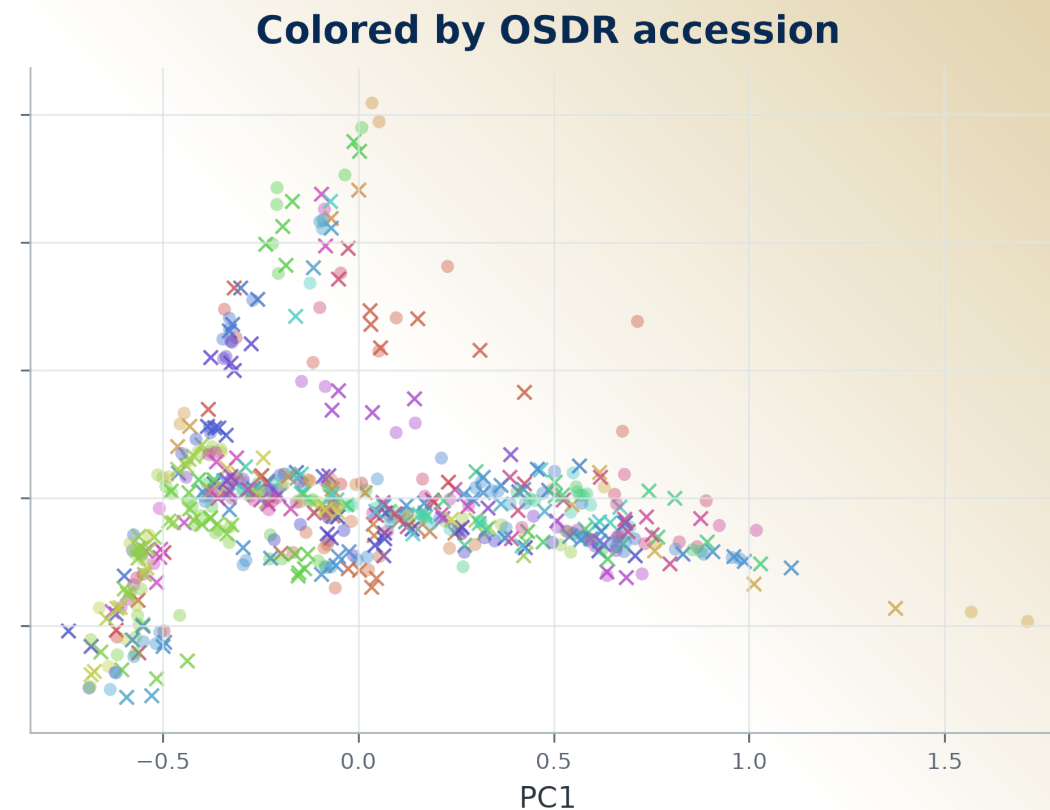
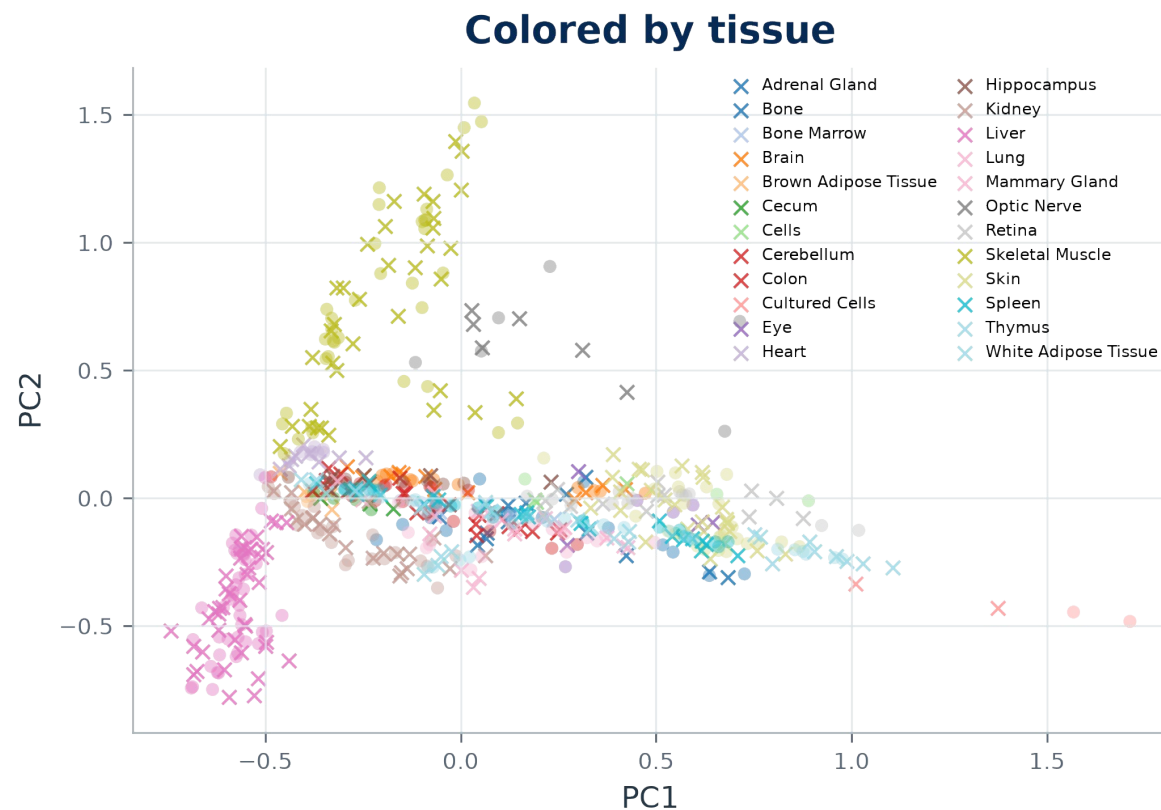
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Tissue and study structure dominate the PCA space

Left: tissue. Right: OSDR accession. Circles are observed profiles; crosses are matched DDIM profiles.

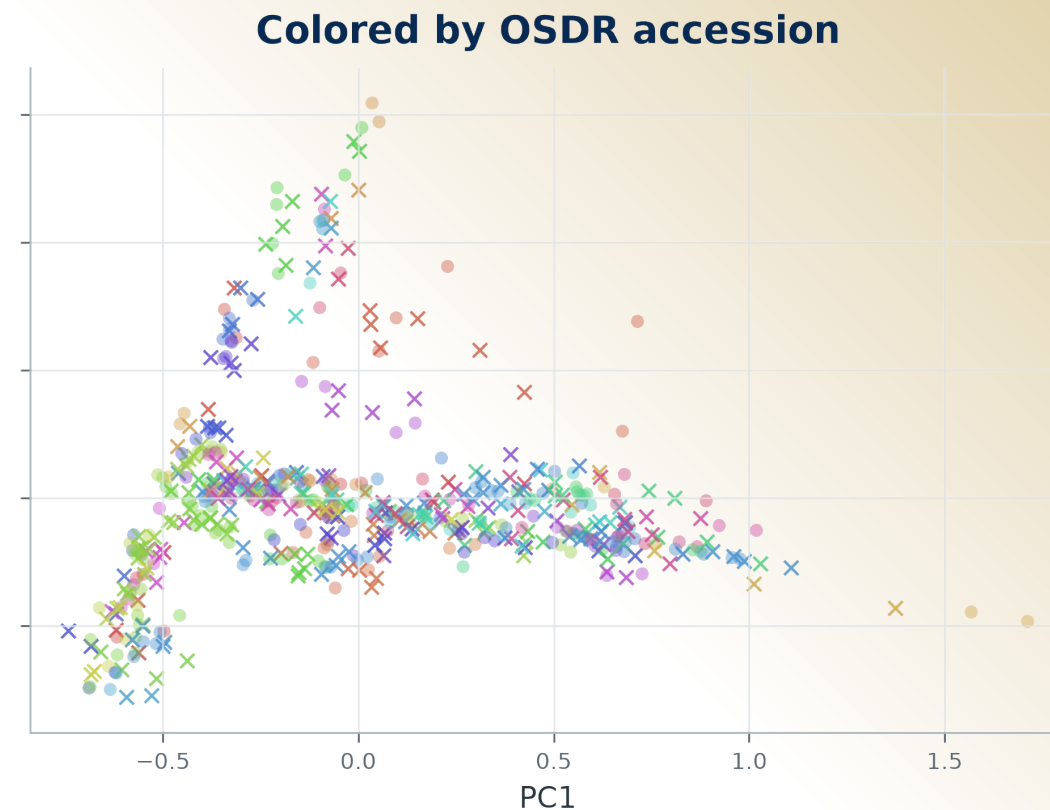
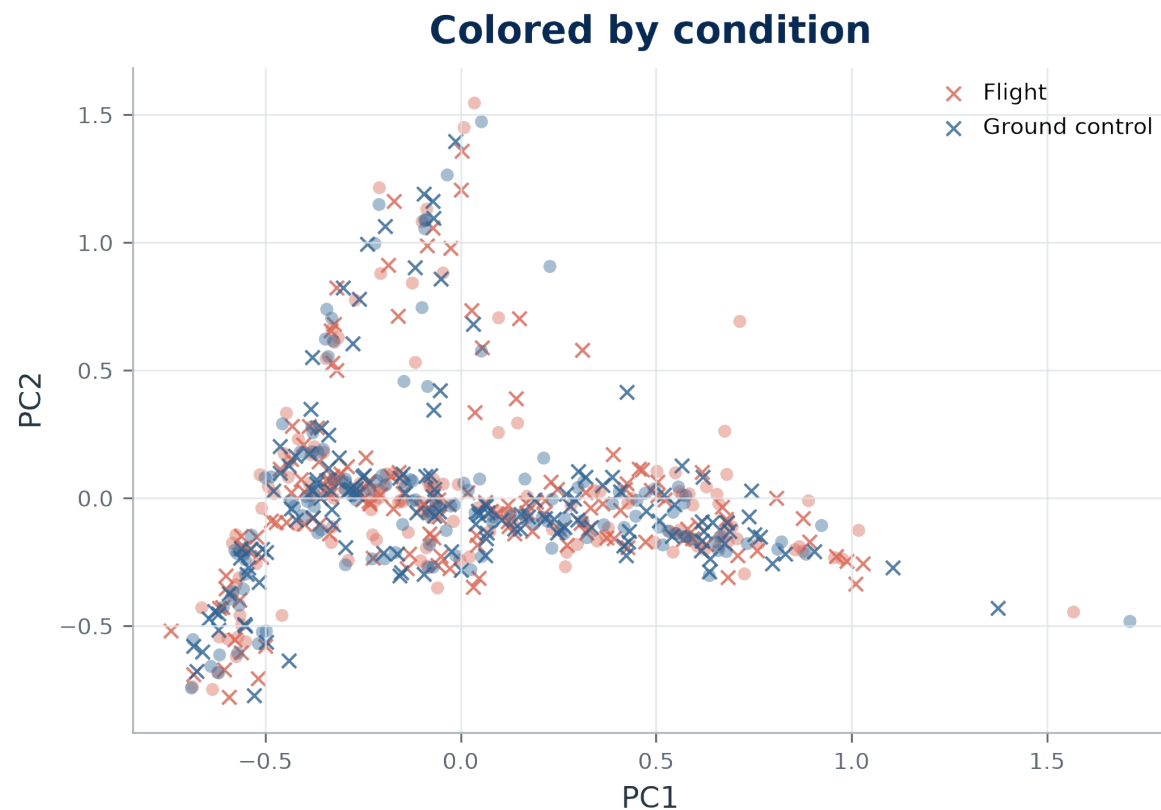


Tissue and accession clusters are visible across both observed and generated profiles.

DIFFUSION OUTPUT

Flight condition is subtler than study structure

Left: FLT or GC. Right: OSDR accession. Circles are observed profiles; crosses are matched DDIM profiles.

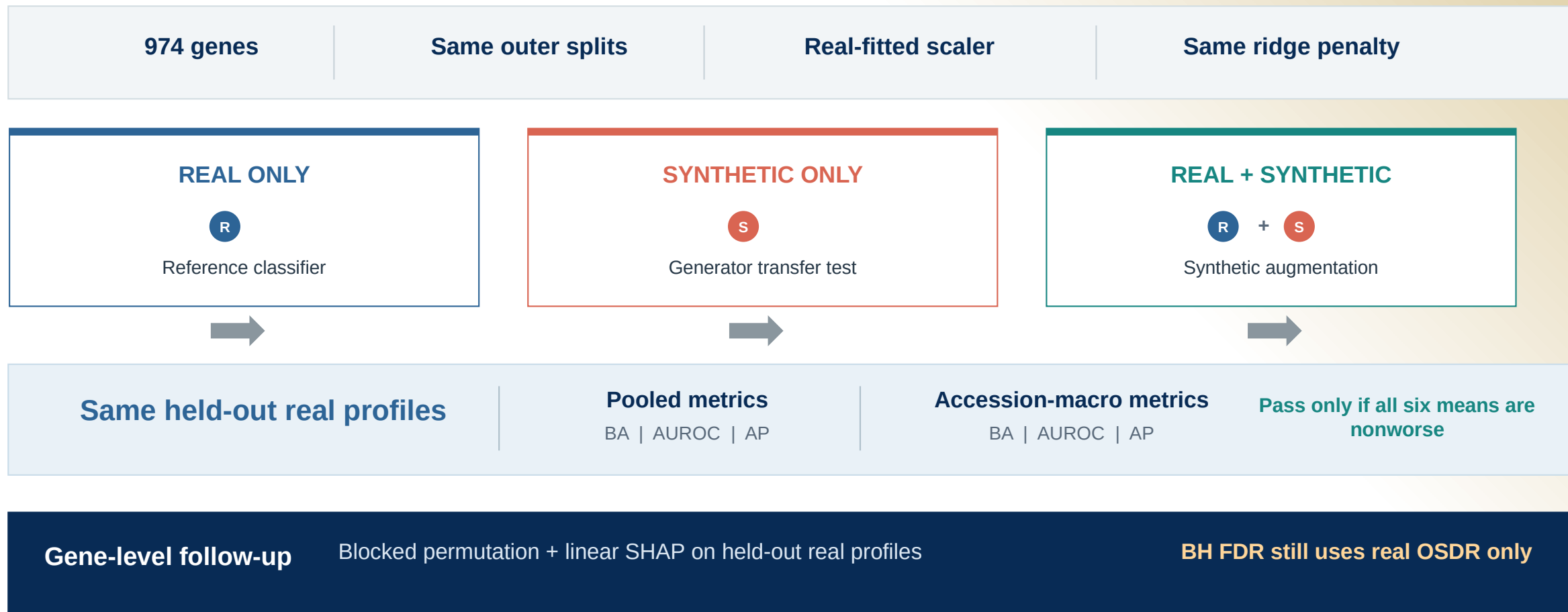


Flight and ground-control profiles overlap within the larger accession-defined structure.

PRIMARY ANALYSIS

Change the training source; hold the classifier fixed

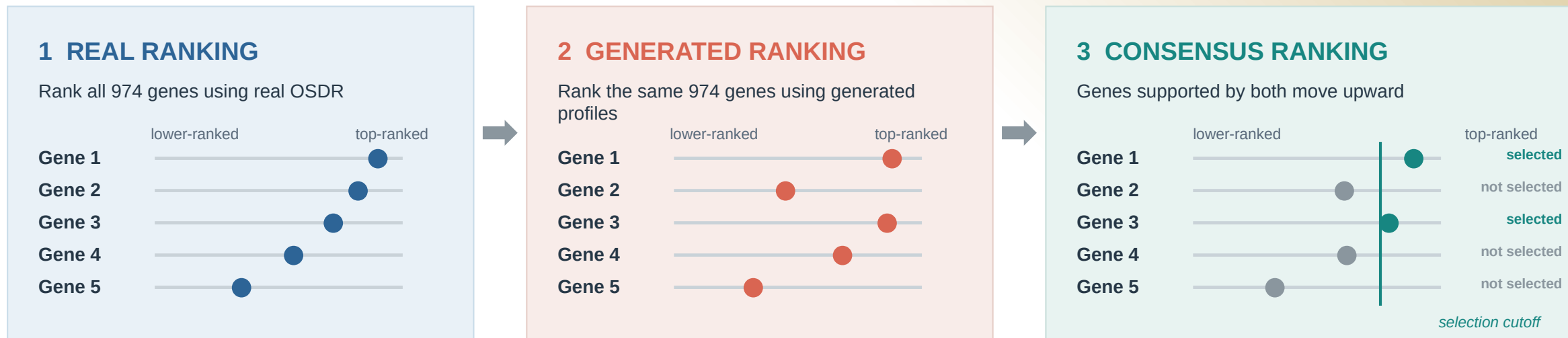
All arms use the same 974 genes, data splits, preprocessing, and ridge regularization.



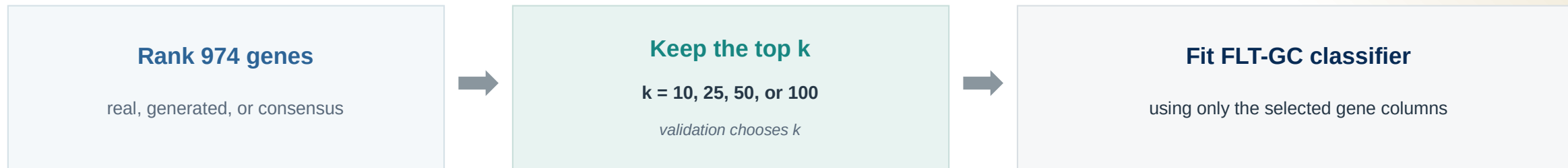
ANALYSIS

Consensus ranking chooses the classifier input genes

Real and generated rankings combine, then only the top k genes are used for FLT-GC classification.



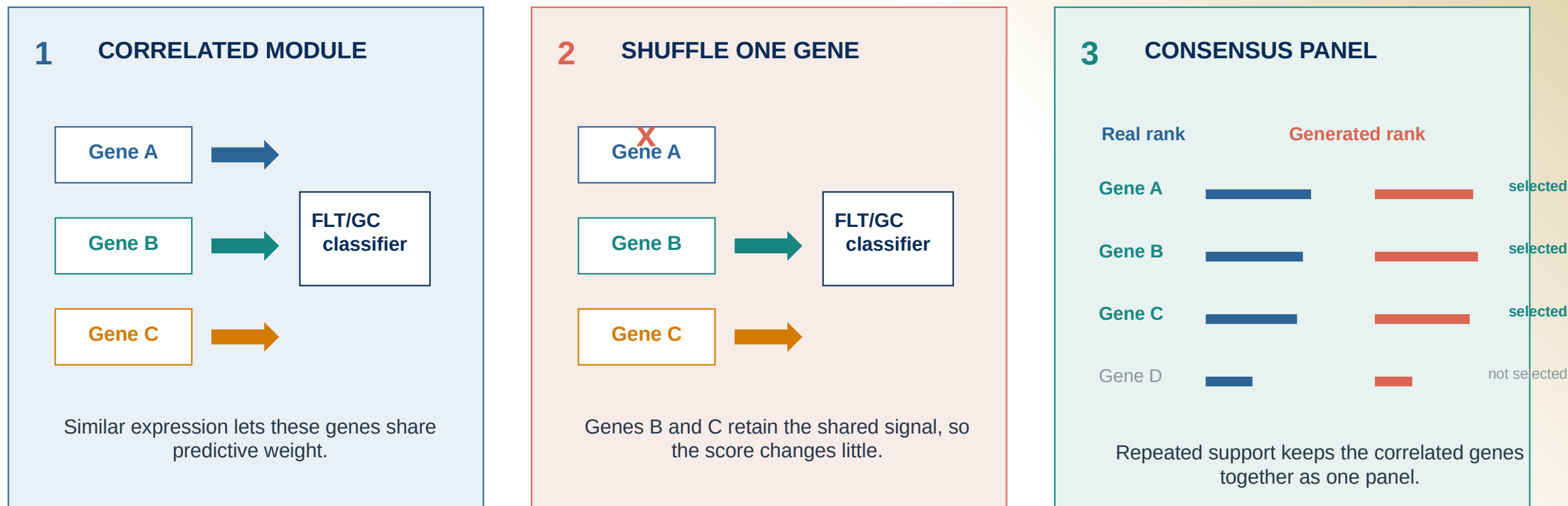
Top-ranked genes become classifier inputs



INTERPRETATION

Correlated genes dilute one-at-a-time importance

A pathway can matter even when no single member causes a large permutation loss.

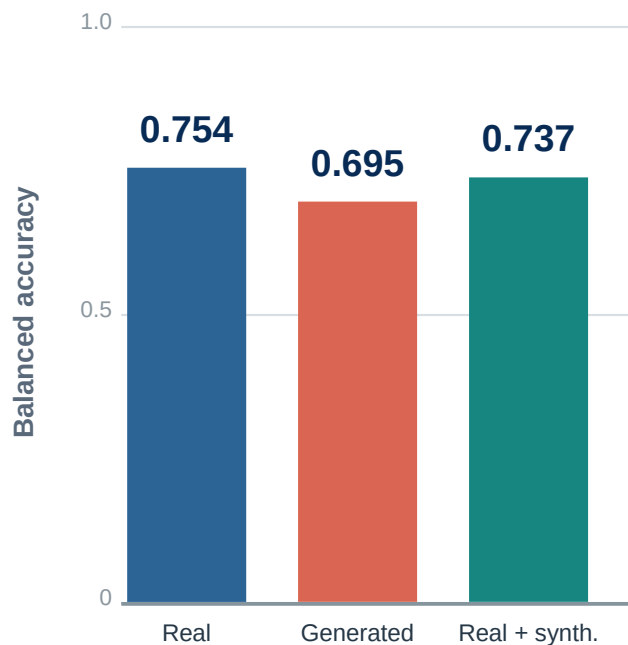


Low marginal importance means a gene was replaceable in this classifier; it does not make the gene or pathway biologically irrelevant.

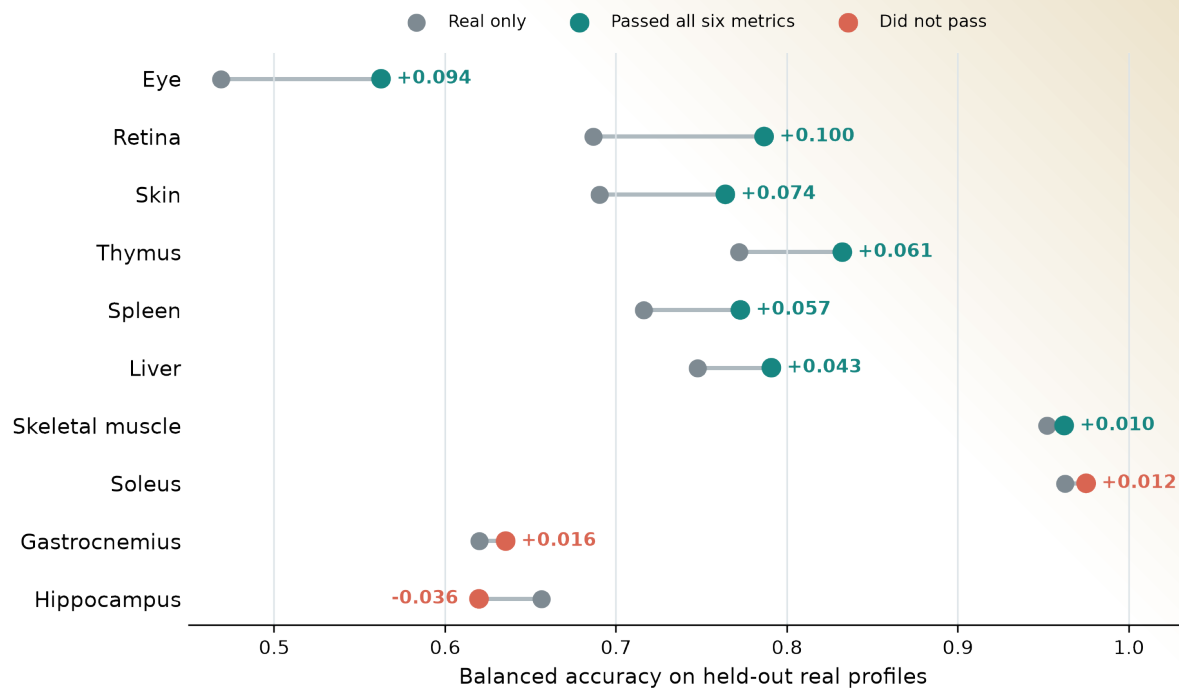
Matched augmentation helped many tissues, but not all

One pooled classifier declined; tissue-specific real-plus-synthetic models were compared with fixed all-gene baselines.

Pooled across tissues



Matched real + synthetic by tissue

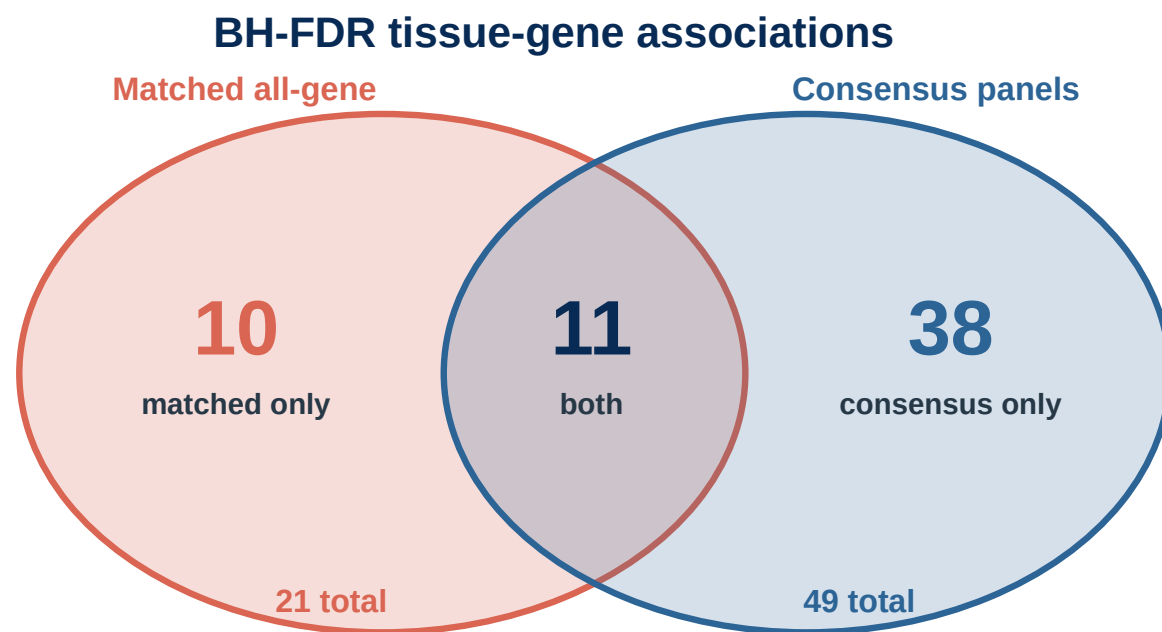


Teal passed all pooled and accession-macro metrics; coral did not.

FEATURE ANALYSIS

Matched and consensus analyses answer different questions

Matched importance tests synthetic contribution; consensus ranking organizes compact correlated panels.

**PRIMARY****Matched all-gene**

Same genes and classifier; asks whether training source changes held-out-real utility or importance.

SECONDARY**Consensus panel**

Ranks compact top-k panels; preserves groups of correlated genes that can replace one another.

SHARED**Real association gate**

FLT-GC effects and BH FDR are estimated from observed OSDR profiles only.

Thymus is strongest because matched importance and the consensus cell-cycle panel converge.

Consensus selection and literature are separate dimensions

All 49 secondary consensus associations received a selection status and an independent literature interpretation.

01	Fix the candidate Gene, tissue, and FLT direction	02	Search Spaceflight and mechanism literature	03	Compare Observed result with published evidence	04	Record Category, rationale, and source IDs
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Selection status

26

Promoted
Stable only with synthetic guidance

23

Reinforced
Stable with and without guidance

Literature interpretation

Class	Count	Meaning
■ Aligning	22	Direct or same-tissue process agreement
■ Complementary	19	Related process or mechanism
■ Ambiguous	4	Mixed or context-dependent evidence
■ Unmatched	4	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

PRIMARY ANALYSIS COVERAGE

The matched screen covered all 27 tissue analyses

Predictive utility and gene-level support are separate outcomes.

4 Matched genes

Utility passed + BH-FDR gene importance

ANALYSIS UNITS

- Liver
- Skin
- Spleen
- Thymus

14 Utility only

All six metrics passed; no retained gene

ANALYSIS UNITS

- EDL
- Adrenal Gland
- Bone Marrow
- Brain
- Brown adipose tissue
- Cerebellum
- Colon
- Eye
- Lung
- Mammary Gland
- Optic Nerve
- Retina
- Skeletal muscle (pooled)
- White adipose tissue

9 Gate not passed

At least one mean metric was worse

ANALYSIS UNITS

- Gastrocnemius
- Quadriceps
- Soleus
- Tibialis Anterior
- Bone
- Cecum
- Heart
- Hippocampus
- Kidney

SECONDARY CONSENSUS INVENTORY

Ten tissue analyses contained consensus-selected genes

All 49 associations passed BH FDR in real OSDR data; consensus ranking affected prioritization, not the statistical test.

Rows: FLT direction + status

■ aligning

■ complementary

■ ambiguous

■ unmatched

Gene color = literature | Non-empty rows shown

Thymus

16 genes

FLT higher	Promoted	<i>Hsd17b11</i> , <i>Etv1</i>
FLT higher	Reinforced	<i>Snx7</i>
FLT lower	Promoted	<i>Nusap1</i> , <i>Stmn1</i> , <i>Birc5</i> , <i>Cdk1</i> , <i>Top2a</i> , <i>Ccnb2</i> , <i>Aurka</i> , <i>Ccne2</i> , <i>Kif20a</i> , <i>Pcna</i> , <i>Ccnf</i>
FLT lower	Reinforced	<i>Ube2c</i> , <i>Gmnn</i>

Spleen

4 genes

FLT higher	Promoted	<i>Rai14</i> , <i>Myl9</i> , <i>Ptprk</i>
FLT higher	Reinforced	<i>Loxl1</i>

Kidney

2 genes

FLT higher	Promoted	<i>Inpp4b</i>
FLT higher	Reinforced	<i>Slc37a4</i>

Gastrocnemius

2 genes

FLT higher	Promoted	<i>Nfkbia</i>
FLT lower	Promoted	<i>Fhl2</i>

Eye

1 gene

FLT lower	Reinforced	<i>Klhl21</i>
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Skeletal muscle (pooled)

12 genes

FLT higher	Reinforced	<i>Sox4</i> , <i>Cebpd</i> , <i>Sh3bp5</i> , <i>Prkcd</i> , <i>Arid5b</i> , <i>Sesn1</i> , <i>Tle1</i>
FLT lower	Promoted	<i>Klhl21</i> , <i>Mapkapk5</i> , <i>Reep5</i> , <i>Itgb5</i>
FLT lower	Reinforced	<i>Bphl</i>

Soleus

5 genes

FLT higher	Reinforced	<i>Tpm1</i>
FLT lower	Reinforced	<i>Bdh1</i> , <i>Ech1</i> , <i>Bnip3</i> , <i>Decr1</i>

Tibialis anterior

4 genes

FLT higher	Promoted	<i>Cebpd</i>
FLT higher	Reinforced	<i>Cdkn1a</i> , <i>St3gal5</i> , <i>Bnip3</i>

Adrenal gland

2 genes

FLT lower	Promoted	<i>Psmb8</i>
FLT lower	Reinforced	<i>Tspan4</i>

Skin

1 gene

FLT higher	Promoted	<i>Plscr1</i>
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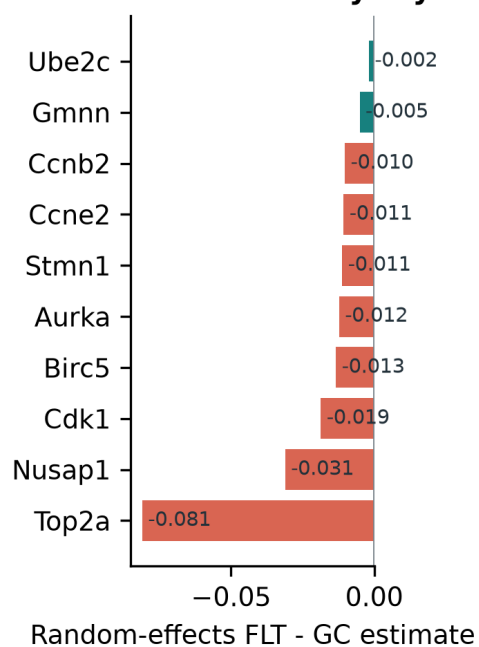
THYMUS

Thymus is strongest across both analyses

Matched importance supports synthetic contribution; consensus ranking expands the correlated cell-cycle panel.

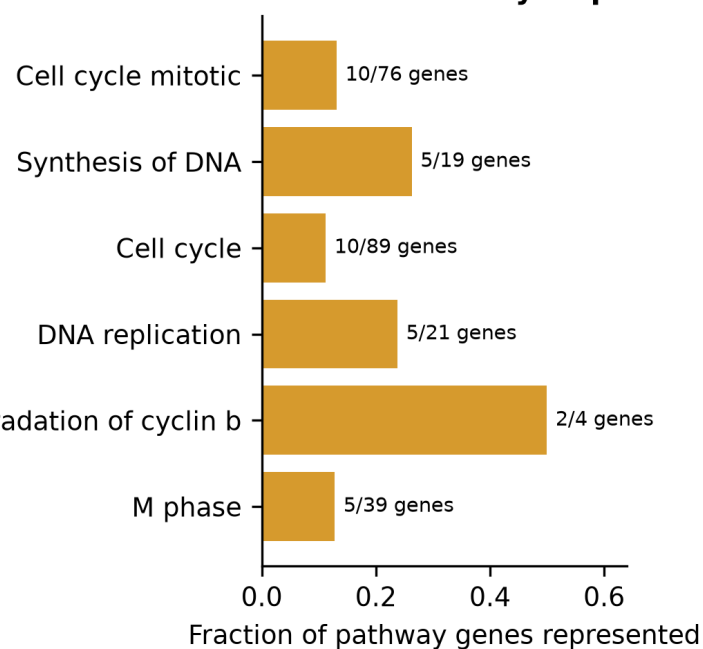
Synthetic-informed thymus genes converge on a flight-lower mitotic program

A Cross-study thymus effects



APC/C CDC20 mediated degradation of cyclin b

B Shared cell-cycle processes



MATCHED ALL-GENE

15

BH-FDR genes with
synthetic-supported
importance

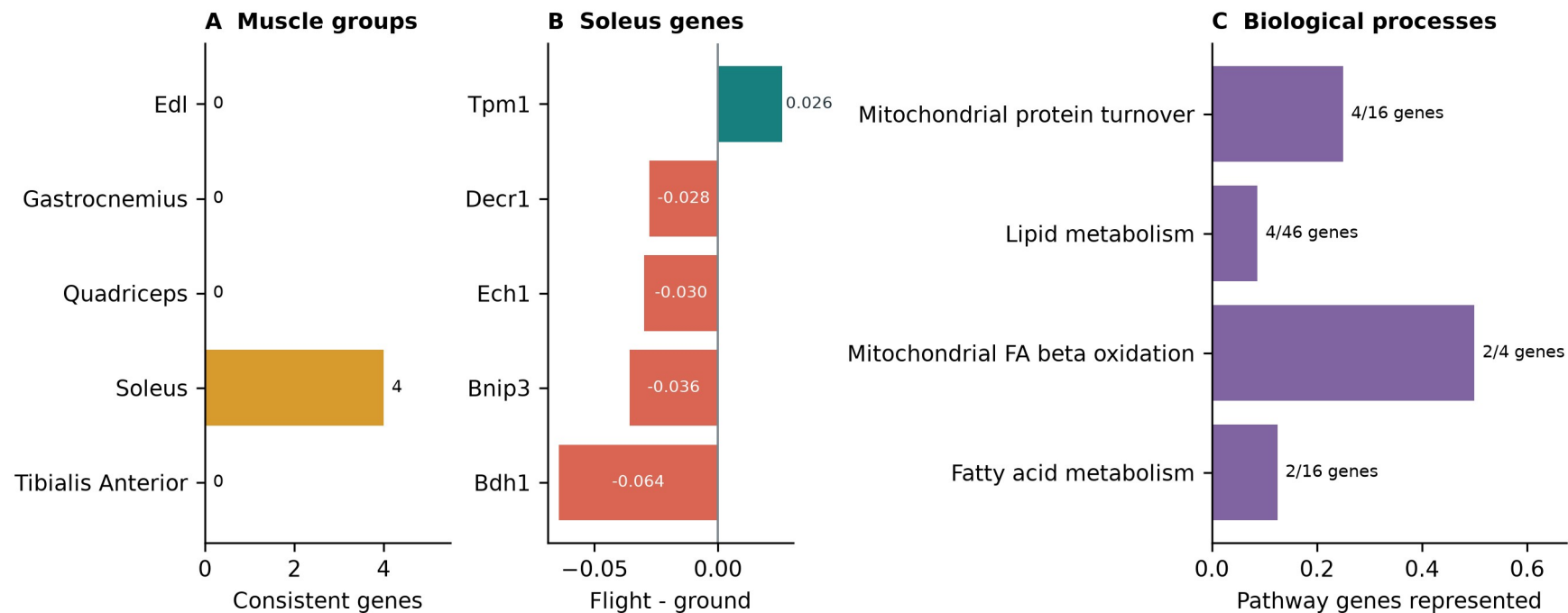
7 promoted | 8 shared

- 26 significant Reactome terms; mitotic cell cycle leads
- Nine genes overlap the 16-gene consensus panel
- Consensus adds correlated Cdk1, Top2a, Aurka, Ccne2, Pcna and Ccnf
- Lower mitotic renewal or fewer cycling thymocytes in flight

Soleus is a coherent secondary consensus result

The five-gene program is biologically interpretable, but the matched all-gene gate did not pass.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



5-gene consensus panel

reinforced in real and consensus rankings

Matched gate not passed

BA +0.013 | AUROC -0.015 | AP -0.009

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Panel-level hypothesis, not matched synthetic contribution

ADDITIONAL MATCHED FINDINGS

Liver, skin and spleen add narrower gene-level results

Pooled muscle improves prediction, while its gene panel remains secondary consensus evidence.

Analysis unit	Evidence	Genes and FLT direction	Interpretation
Liver	Matched	Lower: Grb10, Ppic, H2-DMa, Gtf2a2	Four shared-importance candidates; no significant Reactome set enrichment.
	Consensus	No retained panel	
Skin	Matched	Higher: Plscr1	Matched and consensus support the same gene; broader cell-cycle and DNA-repair context is panel-level.
	Consensus	Higher: Plscr1 plus broader selected pathways	
Spleen	Matched	Higher: Loxl1	Loxl1 is the matched anchor; the four-gene adhesion and cytoskeletal panel is tentative.
	Consensus	Higher: Loxl1, Rai14, Ptprk, Myl9	
Pooled muscle	Matched	All six metrics passed; no gene passed the full gate	Predictive utility is matched; individual gene interpretation comes from the secondary consensus analysis.
	Consensus	12-gene interferon and sialic-acid panel	

ADDITIONAL CONSENSUS FINDINGS

Several narrower panels did not pass the matched gene gate

These remain tissue-specific hypotheses from the secondary ranking analysis.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkb1a higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

TAKEAWAYS

Use matched tests for contribution; consensus for biology

Matched classifiers test utility; consensus ranking explains correlated programs; biological claims still use OSDR profiles.

01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

02 Test

Matched real-plus-synthetic classifiers passed all six metric checks in 18 of 27 tissue analyses and retained 21 BH-FDR associations.

03 Interpret

Consensus ranking recovered broader correlated panels. Thymus was strongest across both analyses; soleus remained secondary.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSD R | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSD R profiles.